market_boost

equity strategy | family: equity-overlay | status: validated | derived from `balanced_leverage`

What this strategy does

Balanced Leverage plus a market position worth half the account

Why it should work. Alpha book earns the drift; the overlay earns the equity risk premium on the same capital.

Design notes (from the strategy's source file).

Market Boost -- Balanced Leverage plus a half-size position in the overall stock market.

Keeps Balanced Leverage's book completely untouched and holds a separate synthetic market-index position of 0.5x trailing NAV beside it (re-sized quarterly, 2bps cost) -- the 130/30-style long-extension structure real funds use to add market exposure without distorting the alpha signal. This is how the equity side clears the S&P 500's return in this sample.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	open_book
beta_overlay	0.5

Performance

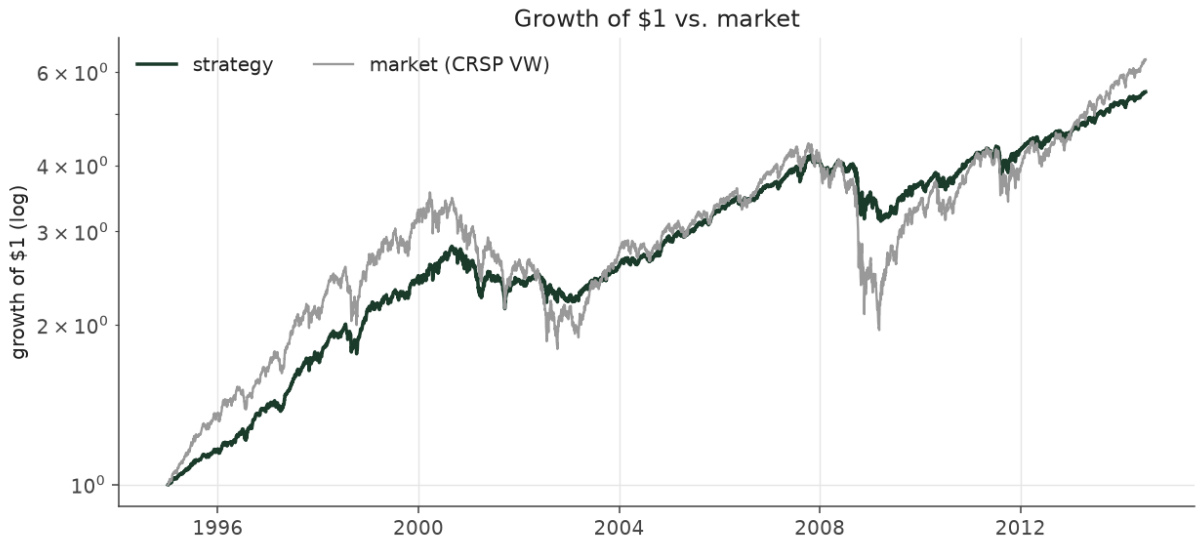
metric	market boost	market (CRSP VW)
Annualized return	9.15%	9.94%
Annualized volatility	10.39%	19.35%
Sharpe (NAV-pct)	0.90	0.59
Sortino	1.29	
Max drawdown	-25.1%	-55.5%
Calmar	0.36	
Final NAV ($10M start)	$55,034,354	
Worst day	-4.84%	
% positive days	55.3%	
Sharpe 1996-2001	1.20	
Sharpe 2002-2007	1.10	
Sharpe 2008-2013	0.40	

Correlation of daily returns with the market: **0.94**.

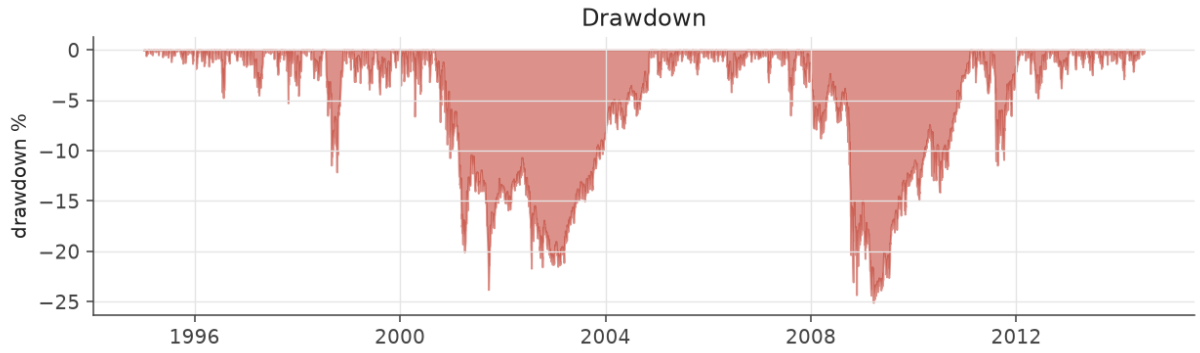
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 0.895

• best return among equity: `market_boost_max` (0.116); this one: 0.0915

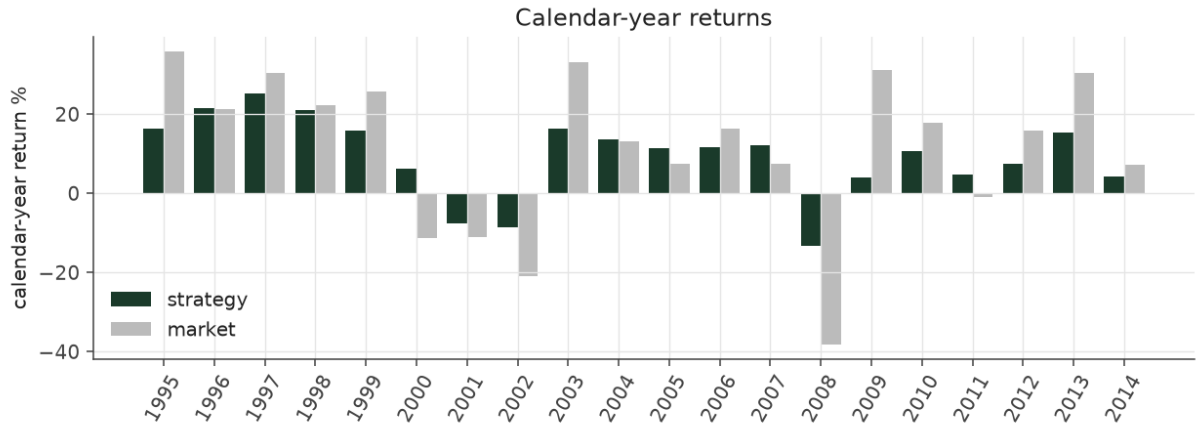
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.365



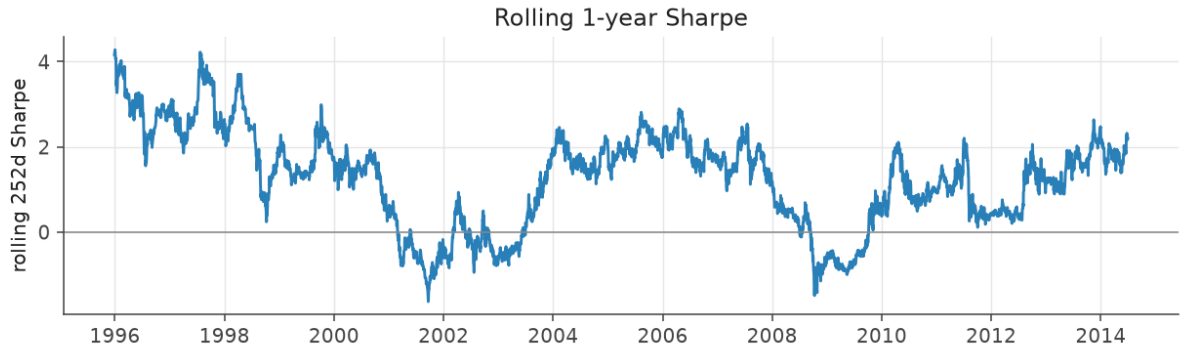
NAV (log scale)



Drawdown from running peak

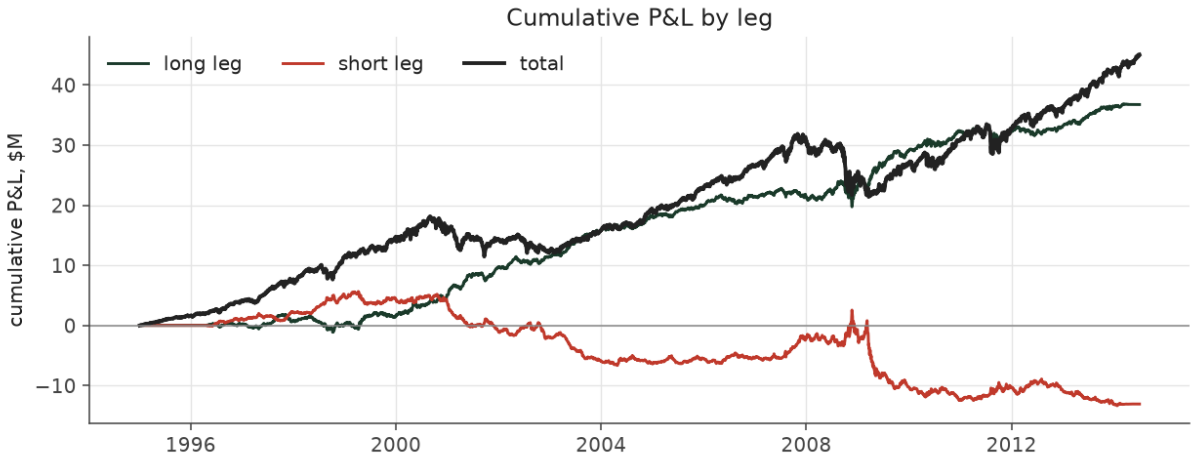


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	36.74	-13.08
P&L after costs ($M)	33.86	-16.13
of which common (average-event) component ($M)	23.41	-24.03
of which excess / PEAD component ($M)	13.34	10.96
excess minus costs ($M)	10.45	7.91
gross return per $ traded (bp)	252.72	-87.98
positions in the money (hit rate)	0.56	0.48
stand-alone Sharpe (leg only)	1.17	-0.32

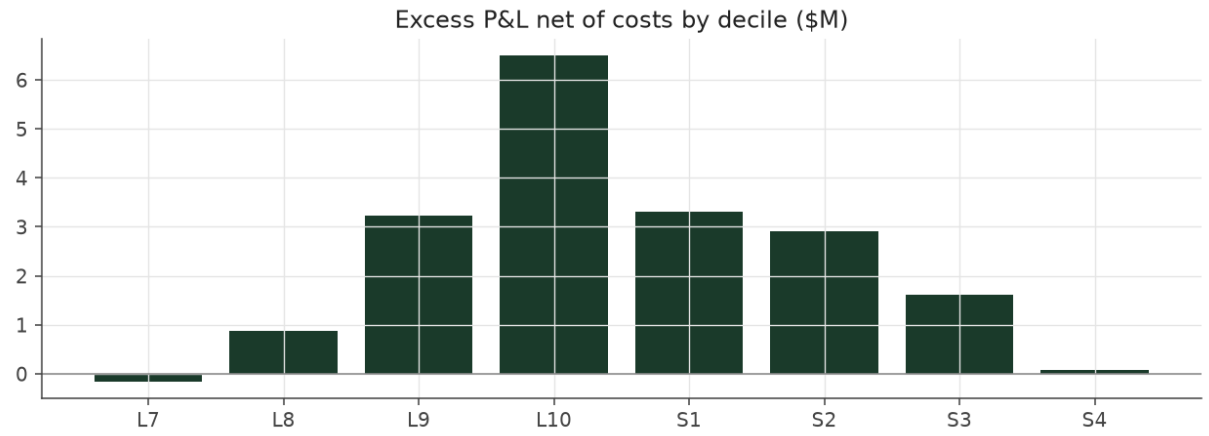
Daily P&L correlation between the two legs: **-0.80**.

• **The short leg loses money in absolute terms** (-13.1M before costs) while the long leg earns +36.7M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 23.4M / 24.0M. Stripping that common component, the surprise signal itself earned +13.3M on the long side and +11.0M on the short side.

• The two legs' daily P&L are -0.80 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.17 long / -0.32 short vs 0.90 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	5,756	95.09	1.46	0.03	0.19	-0.17	-0.9%	-17.41
long	8	15,259	315.71	6.61	1.51	0.63	0.88	4.8%	27.90
long	9	17,582	447.52	9.52	4.12	0.88	3.24	17.6%	72.33
long	10	19,546	595.59	16.27	7.68	1.18	6.50	35.4%	109.09
short	1	17,711	595.10	-6.77	4.58	1.27	3.31	18.0%	55.59
short	2	17,925	511.12	-5.46	3.94	1.03	2.91	15.8%	56.88
short	3	14,022	338.63	-3.40	2.29	0.68	1.62	8.8%	47.74
short	4	2,169	41.38	-0.50	0.15	0.08	0.08	0.4%	18.55



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,094	277.14	0.12	1.83	1.06	0.77	4.2%	27.79
2	21,037	558.41	6.62	8.55	1.56	6.98	38.0%	125.07
3	23,664	672.81	5.00	6.42	1.41	5.01	27.3%	74.49
4	25,541	712.39	3.70	4.61	1.11	3.49	19.0%	49.05
5	26,634	719.38	2.29	2.89	0.79	2.10	11.4%	29.18

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	12,373	201.26	2.28	3.67	1.01	2.67	14.5%	132.52
2	21,374	594.43	4.04	6.62	1.78	4.84	26.4%	81.40
3	23,903	716.16	5.43	7.18	1.43	5.75	31.3%	80.22
4	25,542	717.00	3.84	4.39	1.00	3.39	18.4%	47.22

5	26,774	711.11	2.13	2.42	0.71	1.70	9.3%	23.97

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	38,892	1,003.09	0.23	7.87	1.91	5.96	32.5%	59.44
41-60d	26,124	736.12	2.12	5.39	1.61	3.78	20.6%	51.39
61-90d	35,980	971.10	9.61	6.14	1.78	4.35	23.7%	44.84
>90d	8,974	229.82	5.76	4.90	0.64	4.26	23.2%	185.34

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	32,082	648.44	7.56	9.08	1.28	7.81	42.5%	120.38
2002-07	36,273	996.05	8.23	10.09	1.97	8.12	44.2%	81.55
2008-13	41,615	1,295.64	1.94	5.12	2.69	2.43	13.2%	18.75

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	19,612	276.98	0.79	1.35	0.58	0.77	4.2%	27.70
Chems	2,545	200.33	0.95	1.33	0.36	0.97	5.3%	48.46
Durbl	2,812	217.25	1.56	2.10	0.40	1.70	9.2%	78.15
Enrgy	6,250	246.54	3.05	3.55	0.47	3.08	16.8%	124.98
Hlth	10,936	280.00	1.24	1.86	0.59	1.27	6.9%	45.36
Manuf	11,449	262.67	1.40	2.01	0.55	1.46	7.9%	55.51
Money	15,175	240.41	1.92	2.55	0.55	2.00	10.9%	83.10
NoDur	4,887	242.43	1.08	1.49	0.47	1.02	5.6%	42.24
Other	16,350	265.41	1.89	2.56	0.59	1.97	10.7%	74.26
Shops	11,320	275.81	2.69	3.28	0.57	2.70	14.7%	98.05
Telcm	3,942	242.50	1.36	1.86	0.45	1.41	7.7%	58.27
Utils	4,692	189.81	-0.21	0.36	0.35	0.00	0.0%	0.20

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	36,794	943.20	1.67	4.44	1.72	2.73	14.9%	28.92
low vol	36,578	998.17	8.19	10.51	2.22	8.29	45.2%	83.09
mid vol	34,866	948.23	7.19	8.60	1.90	6.70	36.5%	70.69

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	9.15%	-0.00%
0.5%	1.5	9.00%	0.15%
1.0%	3.0	8.84%	0.31%
2.0%	6.0	8.51%	0.64%
5.0%	15.0	7.39%	1.77%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.